Multi-Agent AI-Based QoE Analysis for 5G Networks using LangChain and LangGraph

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1. Model Setup

1.1 Dependencies and Ollama Installation

For this experiment, the required Python dependencies were installed using pip, including:

- langchain
- langchain-ollama
- langgraph
- langchain-core

Ollama was downloaded and installed from the official website.
After installation, the required local LLM model was pulled using the following command:

```
ollama pull llama3.1:8b
```

This command successfully downloaded the Llama 3.1 8B model which is used for local inference in this experiment.

```
Ollama model pull

PS C:\Users\...\ip\lab4> ollama pull llama3.1:8b
pulling manifest
pulling 667b0c1932bc: 100% |▕▕▕▕▕▕▕▕▕▕▕▕| 4.9 GB
pulling 948af2743fc7: 100%
pulling 0ba8f0e314b4: 100%
pulling 56bb8bd477a5: 100%
pulling 455f34728c9a: 100%
verifying sha256 digest
writing manifest
success
```

1.2 Model Configuration

The system uses the Llama 3.1 8B model which runs locally using Ollama.

```
ChatOllama(model="llama3.1:8b", temperature=0)
```

Temperature is set to 0 to ensure deterministic and consistent output during classification.

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2. System Implementation

2.1 Agents Implemented

The system consists of multiple specialized agents that work together in sequence:

1. **KPI Sanity Check Agent**
   Validates whether the given network KPI values fall within acceptable physical ranges:
   - RSSI (-40 dBm to -140 dBm)
   - Latency (> 0 ms)
   - Packet Loss (0% to 100%)

- Network Type (4G, 5G NSA, 5G SA)
  Outputs VALID or INVALID with reason.

2. **QoE Classification Agent**
   Classifies the user experience into one of the following categories:
   - Smooth
   - Acceptable
   - Poor
     based on latency and packet loss using few-shot learning.
3. **Human Review Agent**
   Allows manual verification of classification before proceeding further.
   In this implementation, it auto-approves for workflow continuity.
4. **Network Advice Agent**
   Generates user-friendly network improvement advice based on QoE classification.
5. **Memory Agent**
   Stores interaction history including QoE classification and generated advice.
6. **Trend Analysis Agent**
   Analyzes historical memory to detect performance trends and provide long-term recommendations.

2.2 LangGraph Workflow

The workflow was built using a state-based architecture with `StateGraph(QoEState)`.

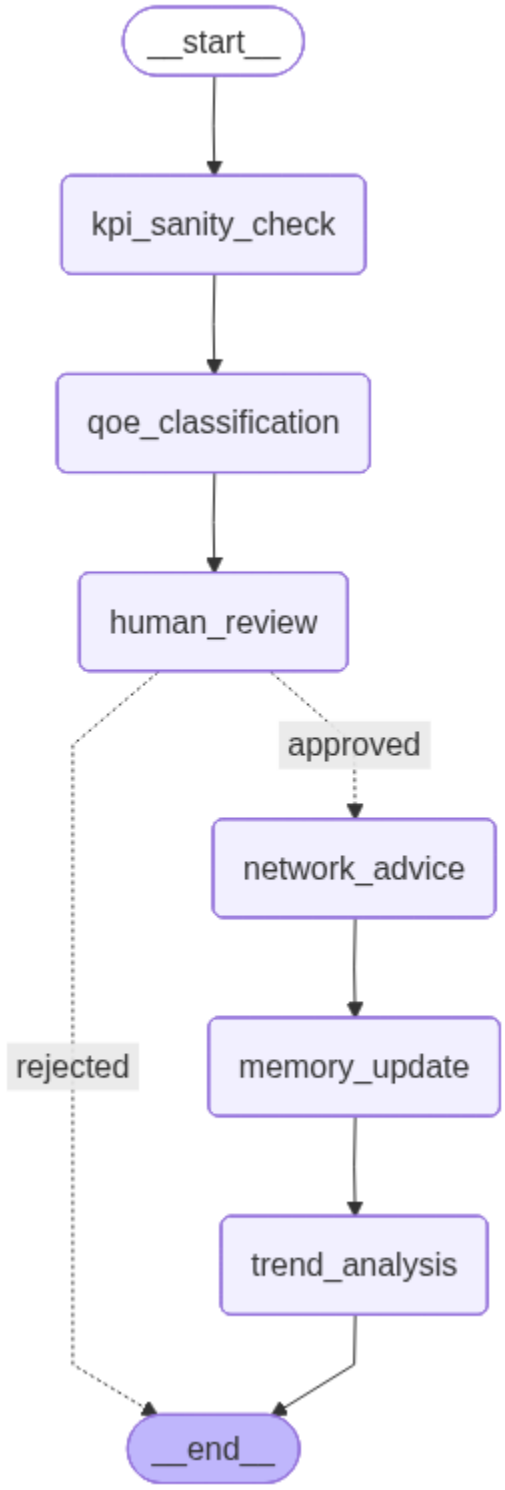
Agent sequence:

```
kpi_sanity_check → qoe_classification → human_review
→ network_advice → memory_update → trend_analysis → END
```

Conditional routing is implemented after the human review step:

- Approved → Continue workflow

- Rejected → End workflow



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3. Experimental Tasks

3.1 Modified Few-Shot Examples

Additional few-shot examples were added to improve classification accuracy in the latency range of 80–120 ms:

- 85 ms, 1.8% → Acceptable
- 90 ms, 2.2% → Acceptable
- 100 ms, 2.8% → Poor
- 110 ms, 2.0% → Poor
- 95 ms, 2.5% → Acceptable

This helped the model better distinguish between Acceptable and Poor QoE.

3.2 Enhanced Network Advice Agent

The Network Advice Agent was enhanced to consider:

- RSSI strength
  If RSSI < -100 dBm, suggest improving signal strength.
- Network Type
  If 4G or 5G NSA is used, suggest upgrading to 5G SA when available.
- Time of Day
  If latency > 80 ms, suggest possible network congestion and recommend trying off-peak hours.

3.3 Real Human Review Implementation

The Human Review Agent was modified to request manual approval from the user using:

```
Approve this classification? (yes/no)
```

If rejected, the workflow terminates without generating network advice.

3.4 Trend Analysis Agent

The Trend Analysis Agent:

- Reads stored QoE results from memory
- Detects patterns such as consistently poor performance
- Provides long-term network improvement suggestions such as:
    - Changing operator
    - Upgrading plan
    - Changing network type
    - Relocating for better signal

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4. Scenarios and Outputs

Scenario A: Excellent Connection

Input:

- RSSI: -65 dBm
- Latency: 12 ms
- Packet Loss: 0.1%
- Network Type: 5G SA

```
Scenario A: Excellent Connection

============================================================
5G NETWORK QUALITY ANALYSIS REPORT
============================================================

KPI SANITY CHECK
VALID

QoE CLASSIFICATION
Smooth

NETWORK ADVICE
Your experience is smooth! That's great news.
Just a minor suggestion: considering your RSSI is around -65 dBm,
you might get an even better signal if you move to a slightly
different location or adjust the antenna.

TREND ANALYSIS
* Observed trend: No significant changes in QoE over time.
* Long-term recommendation: Consider adjusting the antenna
  or moving to optimize signal strength (RSSI).

FINAL VERDICT
User experience is smooth and satisfactory.
============================================================
```

Scenario B: Moderate Performance

Input:

- RSSI: -95 dBm
- Latency: 55 ms
- Packet Loss: 1.8%
- Network Type: 5G NSA

```
Scenario B: Moderate Performance

============================================================
5G NETWORK QUALITY ANALYSIS REPORT
============================================================

KPI SANITY CHECK
VALID

QoE CLASSIFICATION
Smooth

NETWORK ADVICE
Your experience is smooth, which is great! However, your signal
strength could be improved. Try moving to a location with a
stronger signal. Also, since you're on 5G NSA, when available,
upgrading to 5G SA might provide even lower latency.

TREND ANALYSIS
* Observed trend: Your QoE has been consistently Smooth.
* Long-term: Consider adjusting antenna; upgrade to 5G SA.

FINAL VERDICT
User experience is smooth and satisfactory.
============================================================
```

Scenario C: Invalid Input

Input:

- RSSI: -200 dBm
- Latency: -5 ms
- Packet Loss: 150%
- Network Type: 6G

Scenario C: Invalid Input

```
============================================================
5G NETWORK QUALITY ANALYSIS REPORT
============================================================

KPI SANITY CHECK
INVALID
Invalid RSSI value (-200 dBm) is outside allowed range.

QoE CLASSIFICATION
N/A (Invalid KPIs)

NETWORK ADVICE
It looks like your network connection is not working properly.
RSSI strength is extremely low at -200 dBm. Network type 6G
may be an error. Packet loss of 150% indicates significant
connectivity issues. Contact your operator for assistance.

TREND ANALYSIS
* Observed trend: Poor QoE due to weak signal, high packet loss.
* Long-term: Contact operator for targeted support.

FINAL VERDICT
Cannot determine — KPIs were invalid.
============================================================
```

Final Output Report

Final Output Report (Poor scenario)

```
============================================================
5G NETWORK QUALITY ANALYSIS REPORT
============================================================

KPI SANITY CHECK
VALID

QoE CLASSIFICATION
Poor

NETWORK ADVICE
Your experience is poor due to the weak signal and high latency.
First, try moving to a location with a stronger signal. Your
current RSSI of -110 dBm is quite low. Also, consider upgrading
to 5G SA when available. The high latency could be due to
congestion — try off-peak hours (early morning or late night).

TREND ANALYSIS
* Observed trend: Poor QoE due to weak signal and high latency.
* Long-term: Upgrade to 5G SA; move to stronger signal location.

FINAL VERDICT
User experience is currently NOT satisfactory.
============================================================
```

- KPI Validation
- QoE Classification
- Network Advice
- Trend Analysis
- Final Verdict

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5. LangSmith Monitoring (Optional)

Insert Screenshot: LangSmith Execution Trace

Observations:

- Workflow followed expected execution path
- QoE classification agent consumed most tokens
- Network advice generation took the longest time
- Overall classification accuracy improved after few-shot modification

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6. References

- 5G QoE Analysis Lab Manual
- LangChain Documentation
- LangGraph Documentation
- Ollama Documentation